

NURI IMRENDI

Computer Engineer

ni.imrendi@gmail.com | 0554 690 5390 | github.com/imndllnuri | linkedin.com/in/nuri-imrendi

Graduating Computer Engineer with hands-on experience in computer architecture research (Barcelona Supercomputing Center), production software engineering (DigiGarden), and personal projects spanning CPU emulation, RTOS/firmware, and FPGA digital design (Vivado, VHDL). Looking for a full-time role in embedded systems, FPGA/RTL, or computer-architecture engineering, working close to the hardware in C/C++.

Education

Abdullah Gül University – Bachelor of Computer Engineering, Kayseri 2019 – 2026

GPA: 3.0/4.0 | Relevant Coursework: Digital Design, Embedded Systems, Computer Organization, Parallel Architecture, Operating Systems, Computer Networks, Internet of Things

Experience

Barcelona Supercomputing Center (BSC)

Research Intern, Computer Architecture

Jul 2025 – Feb 2026

Barcelona, Spain

- Designed a trace-driven microarchitectural evaluation pipeline (Intel Pin/SDE → ChampSim) and developed an IP-stride prefetcher with a high-coverage mode, achieving +5.43% IPC and a 5.17% cycle reduction on HPCG.
- Extended ChampSim's trace format to propagate metadata for selective prefetch aggressiveness; co-authored CoRoLa, a paper on control-flow semantics for cross-page prefetching (submission in progress).

DigiGarden

Software Engineering Intern

Feb 2026 – Jun 2026

Teknopark Kayseri, Türkiye

- Built a cross-platform iOS/Android app (React Native, Expo, TypeScript) for Turkish farmers to manage field data, with real-time Firestore sync and CRUD operations across nested field collections.
- Implemented Firebase email/password and Google Sign-In authentication, push notifications via Firebase Cloud Messaging, a field-health scoring heuristic, and a localized (Turkish/English) news feed via the GNews API.

Projects

RV32I CPU Emulator with Interactive GUI – C++, Python, PyQt5, pybind11, RISC-V Assembly *Personal*

Project

- Built a fetch/decode/execute emulator for the RV32I base ISA plus the RV32M multiply/divide extension, CSR instructions, and exceptions, exposed to Python via pybind11, with a PyQt5 GUI for syntax-highlighted editing, single-step execution, and live register/memory/stack/disassembly views.

Fault Injection Framework for RISC-V – C, Python, gem5, RISC-V, SPEC CPU 2017 *Personal Research Project*

- Built a multiprocessing-parallelized fault-injection framework on gem5's AtomicSimpleCPU/RISC-V model, injecting randomized single-bit register faults across timed execution intervals.
- Evaluated system reliability on SPEC CPU 2017 benchmarks, classifying outcomes (silent data corruption, crash, masked) to identify fault patterns relevant to resilient microarchitecture design.

Configurable Cache Policy Simulator – C++, CMake, Intel Pin/SDE, Python

Personal Project

- Built a two-stage toolchain – an Intel Pin-based memory-access tracer and a modular C++ cache simulator with a factory-based, pluggable replacement-policy architecture – with warmup/measurement phases and cycle-aware statistics (hit rate, AMAT).

Automated Plant Watering System – C, STM32, ADC, DMA, GPIO, Proteus

Embedded Systems Project

- Designed an STM32F103-based embedded system (STM32CubeMX, Keil MDK-ARM) that reads soil-moisture and water-level sensors via ADC with DMA, drives an LCD display and pump relay via threshold-based control, and validated functionality through Proteus simulation.

Smart Attendance Automation System – C, RFID, ESP8266, Python, Flask, MySQL, PyQt5 *IoT Coursework*

Project

- Programmed an ESP8266 NodeMCU to read RFID cards via an MFRC522 reader and report to a Flask API backed by MySQL on Google Cloud; built a PyQt5 GUI for instructors to manage classes, filter attendance by date, export to Excel, and visualize statistics.

Skills

Languages: C, C++, Python, TypeScript/JavaScript, VHDL, RISC-V & x86 Assembly, Bash, SQL

Computer Architecture & Simulation: ChampSim, gem5, Intel Pin/SDE, trace-driven microarchitectural simulation, cache replacement policies

FPGA & Digital Design: Xilinx Vivado, VHDL, Basys3, RTL synthesis/implementation flow, FSM design

Embedded Systems: STM32 (CubeMX, HAL, Keil MDK-ARM), ADC/DMA, GPIO, RFID, ESP8266, Arduino, Proteus simulation

Mobile & Web: React Native, Expo, Firebase (Firestore, Auth, Cloud Functions, FCM), Fastify, REST APIs

Databases & Protocols: MySQL, SQLite, REST/HTTP/JSON, UART, SPI, I2C, TCP/IP

Tools: Git, GitHub Actions, CMake, Linux, Bash

Languages (spoken): Turkish (Native), English (C1 Reading/Listening/Speaking, B2 Writing)